



HARVEST HOLD RWANDA

ELIMINATING POST-HARVEST LOSS THROUGH IOT (internet of things and AI)

Empowering Smallholders | Sustainable Cold-Chain | AI Market Intelligence

IDENTITY & STRATEGIC FRAMEWORK



Our Mission

To pioneer zero-waste agricultural supply chains in Rwanda by deploying affordable, IoT-enabled cooling for every community and every individual farmer



Our Vision

To lead Rwanda's youth-driven AgTech revolution, ensuring 100% of the national harvest reaches the market in peak condition.



Core Values

Precision Protection, Digital Equity, and absolute Quality Preservation for rural smallholder families.

| THE POST-HARVEST CRISIS

40%

Yield Loss in
Horticulture

The Invisible Leakage

In Rwanda, 20% to 40% of horticultural produce rots before it reaches the consumer. This isn't just waste—it's an economic disaster that strips \$140M+ annually from the pockets of rural farming families.

THE RURAL COLD-CHAIN GAP and existing problem

- ❗ **Energy Poverty:** 70% of farmers lack grid access for modern high tech refrigeration. To cool their harvests because there are too expensive to access as individuals
- ❖ **Forced Sales:** Farmers sell fast at low prices to avoid rot, losing all market leverage and spoilage of their harvest
 - ❖ They also lack skills of post harvest loss management
 - ❖ They also lack how they can be connected to international market while they produce high quality and fresh fruits and vegetables every season
- 🚛 **Infrastructure Lack:** Decentralized, affordable cold-chain units simply do not exist to avoid post harvest loss



INNOVATION: THE SHCCS SILO

Smart Hybrid Charcoal Cooling

Our student-engineered **SHCCS** unit uses the natural physical process of evaporative cooling across wet charcoal panels. It drops temperatures by **8-15°C** below ambient, preserving i freshness without refrigerants. The smart hybrid charcoal cooling system ,is our answer ,it uses the physical property of evaporative cooling across wet charcoal panel ,dropping temperature by up to 15 degree celcius without the need for high voltage grid power



Low cost : locally sourced materials

Eco friendly : zero chemical refrigerants

IoT : Managed real time

Impact: Extends shelf life of leafy greens and tomatoes from 2 days to over 14 days or more while connecting them to market.

| THE IOT ARCHITECTURE

Full Schematic Diagram

Driven by an **ESP32 DevKit**, the system integrates:

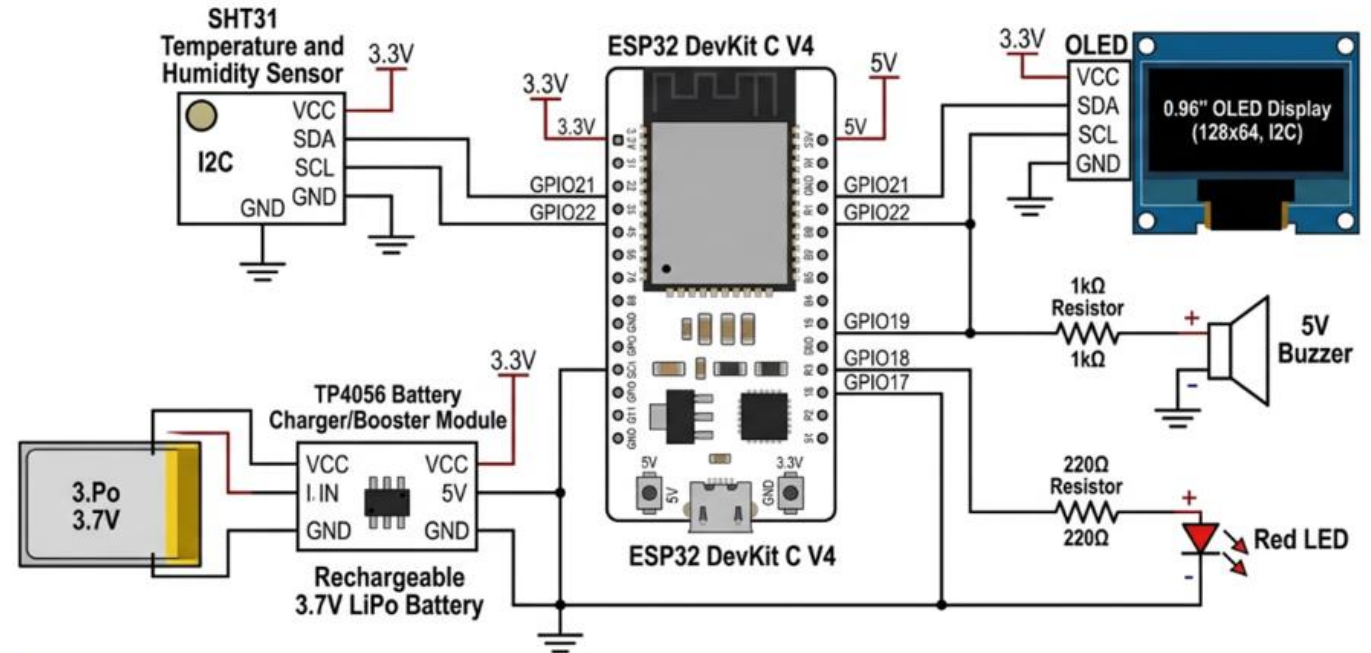
SHT31: High-precision Temp/Humidity.

MQ-131/ Electrochemical: Ethylene sensing to track ripening.

TP4056: Solar-optimized battery management.

OLED Display: Local monitoring for the farmer.

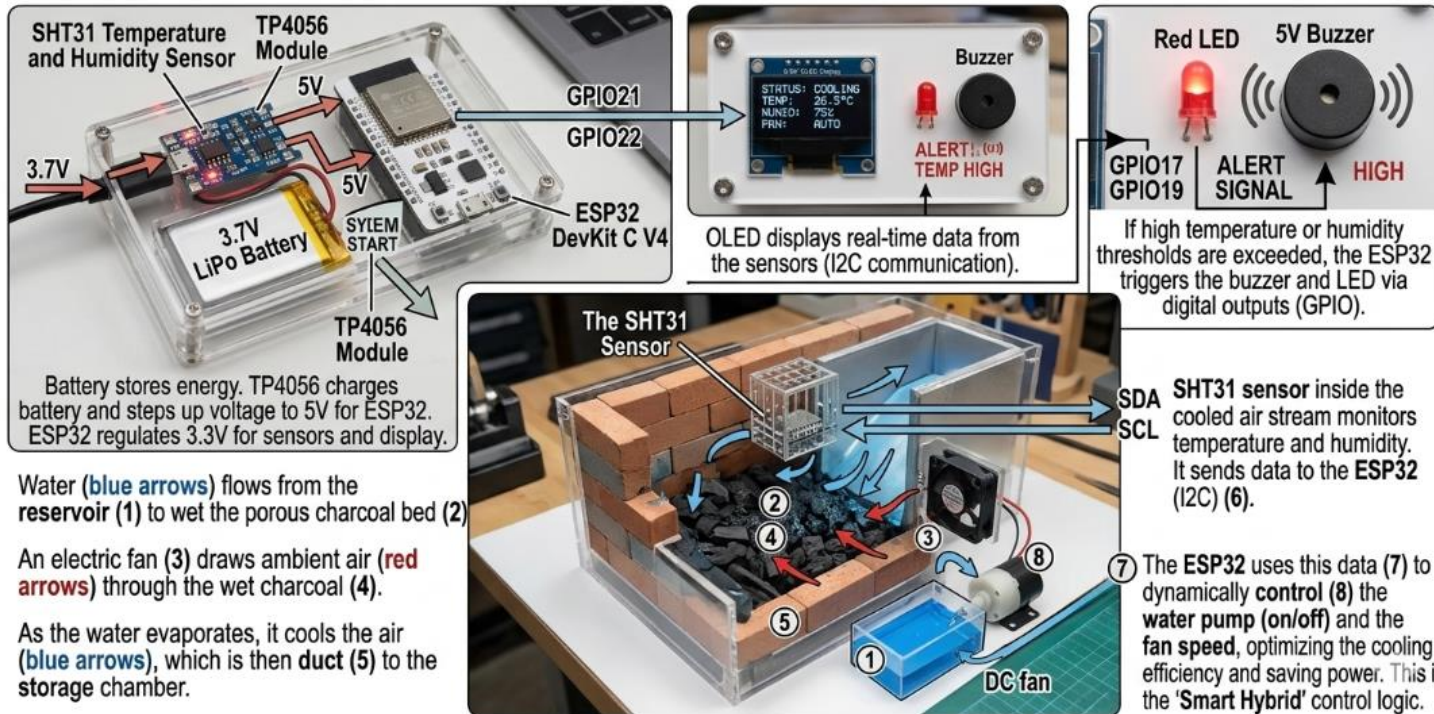
SMART HYBRID CHARCOAL COOLER: FULL SCHEMATIC DIAGRAM



SYSTEM WORKING MECHANISM

The Cooling Cycle

SMART HYBRID CHARCOAL COOLER: PHYSICAL SYSTEM WORKING DIAGRAM



1. **Sensors** detect rising ethylene or temperature.
2. **ESP32** activates solar-powered DC fans.
3. **Water Drip** saturates charcoal panels.
4. **Evaporation** pulls heat from the internal chamber.
5. **Data** is transmitted to our cloud dashboard.

|OFF-GRID FREEDOM

The Solar Advantage

Our project addresses the "Energy Poverty" of Rwandan villages. The SHCCS runs entirely on a **12V Solar PV system**.

By using low-draw DC brushless fans and gravity-fed water irrigation for the charcoal, we ensure that a single 50W panel can keep a whole harvest fresh for weeks.



"CERTIFIED FRESH" STANDARD



Ethylene Monitoring

We don't just cool; we manage ripening. Sensors detect early ethylene release, allowing for early market dispatch.



Data Immutability

Historical storage logs provide proof of quality, giving buyers confidence in the produce's shelf-life.



RSB Alignment

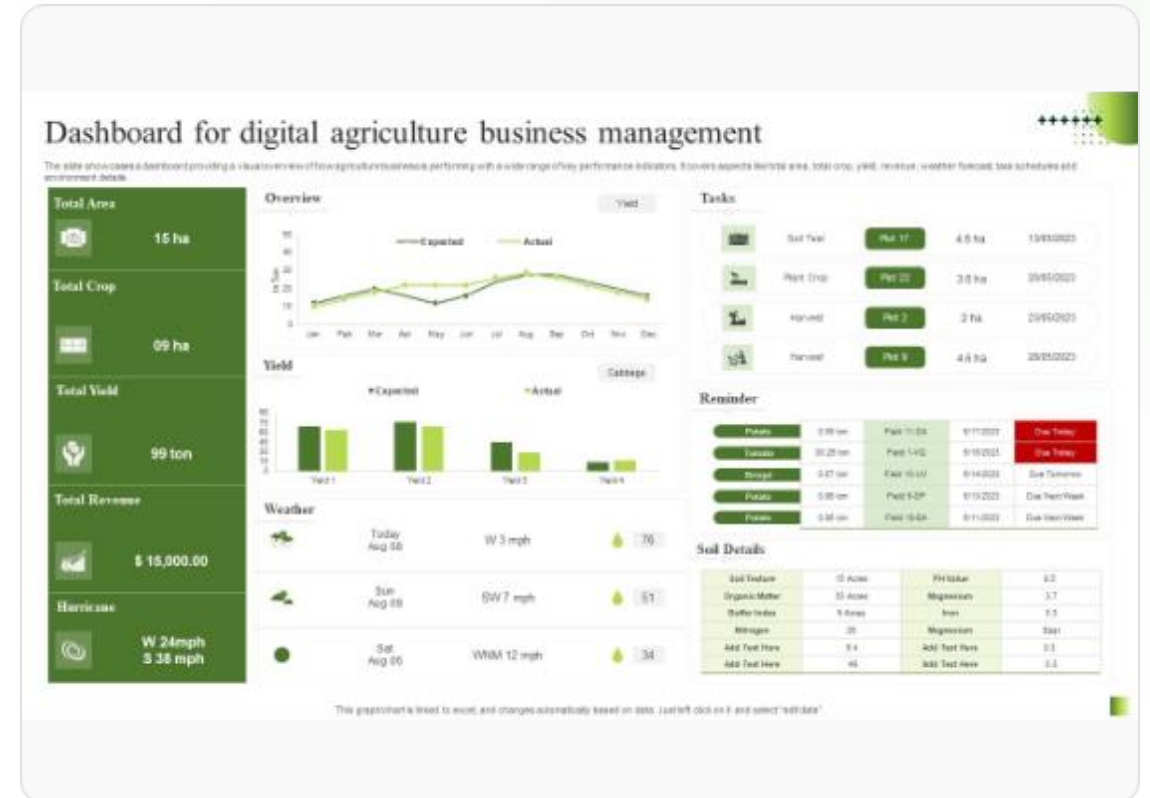
Working toward Rwanda Standards Board certification for IoT-stored produce.

THE DIGITAL MARKETPLACE

AI-Driven Dashboards

Harvest hold Rwanda silos act as **nodes in a network**. When a silo detects that tomatoes are reaching peak ripeness, it automatically lists the inventory on our marketplace.

Exporters, hotels, and wholesalers can "bid" on produce while it is still in the cooler, ensuring a 0-day transit window once removed. Dashboard after controlling temperature and humidity in cold room it also offer virtual certificate ,certify that your vegetables has quality from collaboration from RSB and NAEB for easy for small holder farmer to acces to exportation market and access to diffent market place



| BUSINESS MODEL: SOCIAL ENTERPRISE and how revenue will be maximized

Hardware-as-a-Service (HaaS)

We operate as a social enterprise to ensure scalability. Farmers pay a small "cooling fee" per crate or per day, which covers maintenance and the IoT data subscription and workers. This ensures the project outlives grant cycles and remains a permanent fixture in the agricultural landscape., fees for cooling system ,imagine if we on every market we have cooling hub like this and every farmer pays 100 frw for each crate ,it is amazing but the main focus is impact and changes we want in our society and our nation ,to reduce post harvest loss and securing food security across Rwanda as well as modernizing agriculture

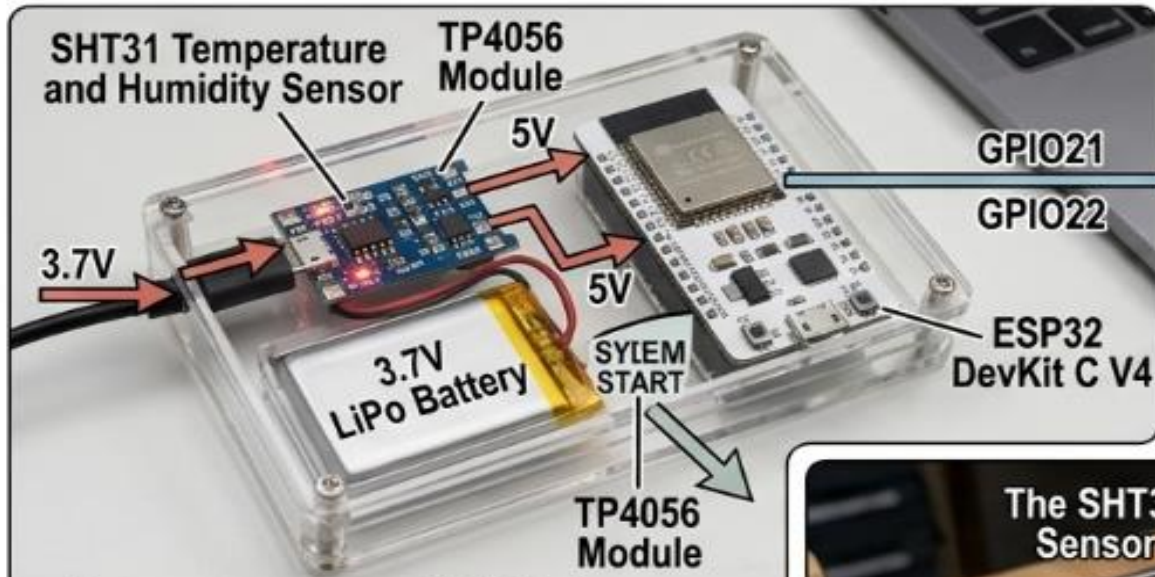
THE HARVEST HOLD RWANDA TEAM

Role	Responsibility
IoT & Hardware Lead	Sensor calibration, ESP32 programming, and SHCCS physical design.
Software Developer	Cloud database management, AI Marketplace, and Mobile UI.
Supply Chain Manager	Partnerships with NAEB, hotels, and logistics providers.
Community Outreach	Farmer training, pilot site selection, and field data collection.

Appendix



SMART HYBRID CHARCOAL COOLER: PHYSICAL SYSTEM WORKING DIAGRAM

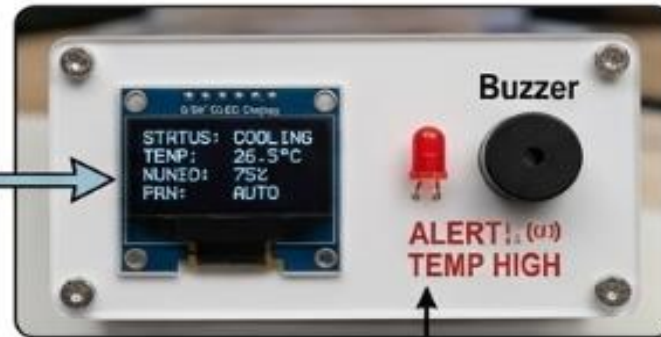


Battery stores energy. TP4056 charges battery and steps up voltage to 5V for ESP32. ESP32 regulates 3.3V for sensors and display.

Water (**blue arrows**) flows from the **reservoir (1)** to wet the porous charcoal bed (**2**)

An electric fan (**3**) draws ambient air (**red arrows**) through the wet charcoal (**4**).

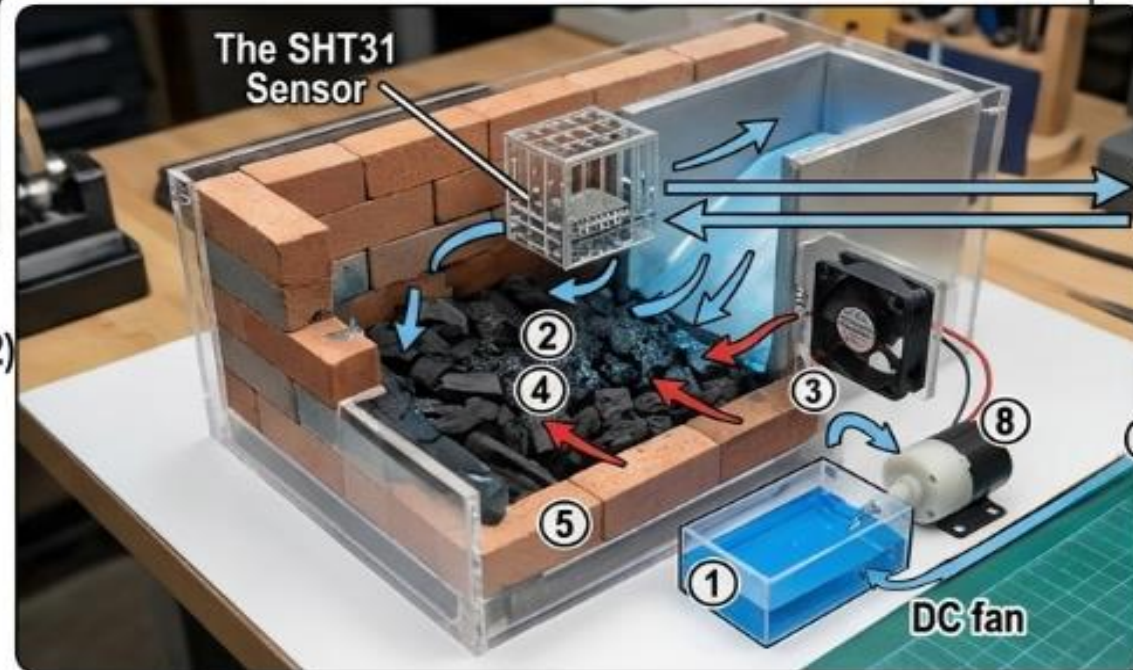
As the water evaporates, it cools the air (**blue arrows**), which is then **duct (5)** to the storage chamber.



OLED displays real-time data from the sensors (I2C communication).



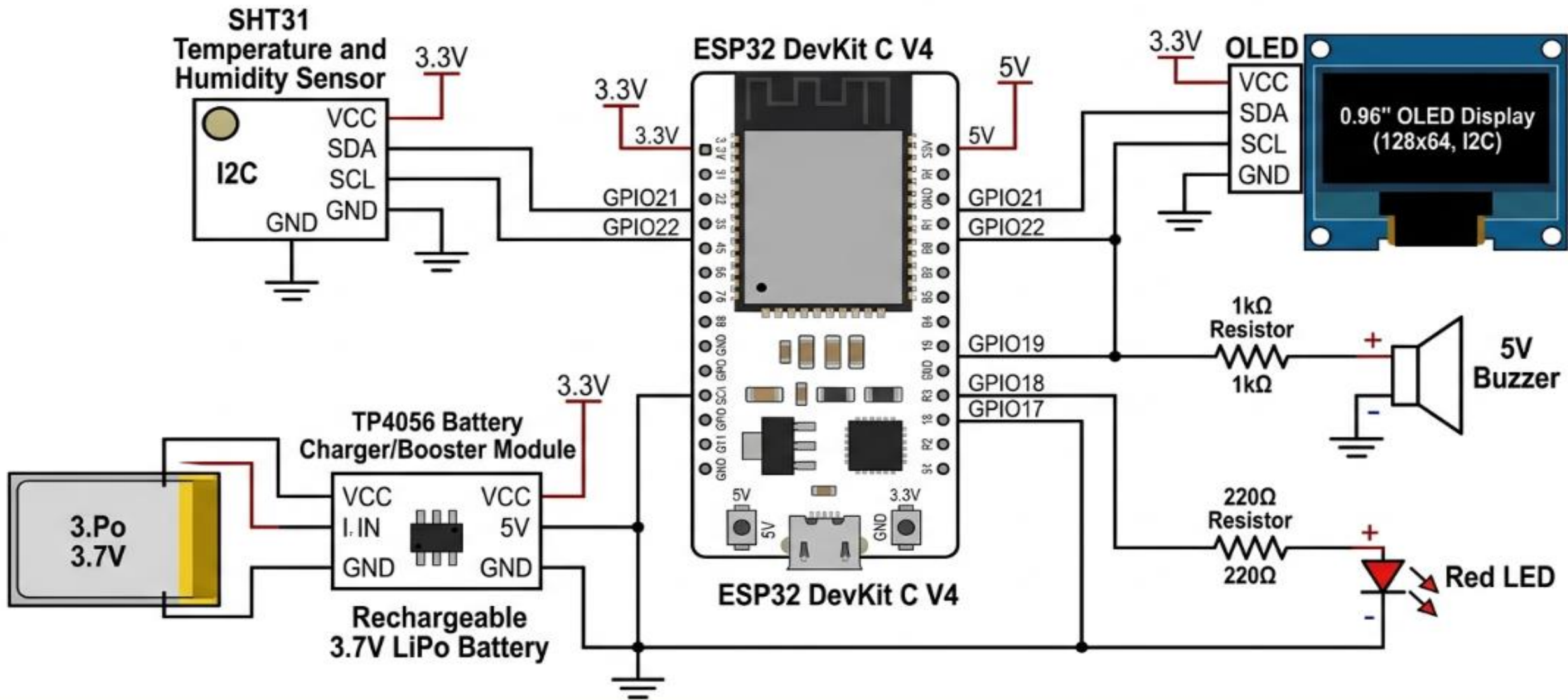
If high temperature or humidity thresholds are exceeded, the ESP32 triggers the buzzer and LED via digital outputs (GPIO).



SHT31 sensor inside the cooled air stream monitors temperature and humidity. It sends data to the **ESP32** (I2C) (**6**).

(7) The **ESP32** uses this data (**7**) to dynamically **control (8)** the **water pump (on/off)** and the **fan speed**, optimizing the cooling efficiency and saving power. This is the 'Smart Hybrid' control logic.

SMART HYBRID CHARCOAL COOLER: FULL SCHEMATIC DIAGRAM

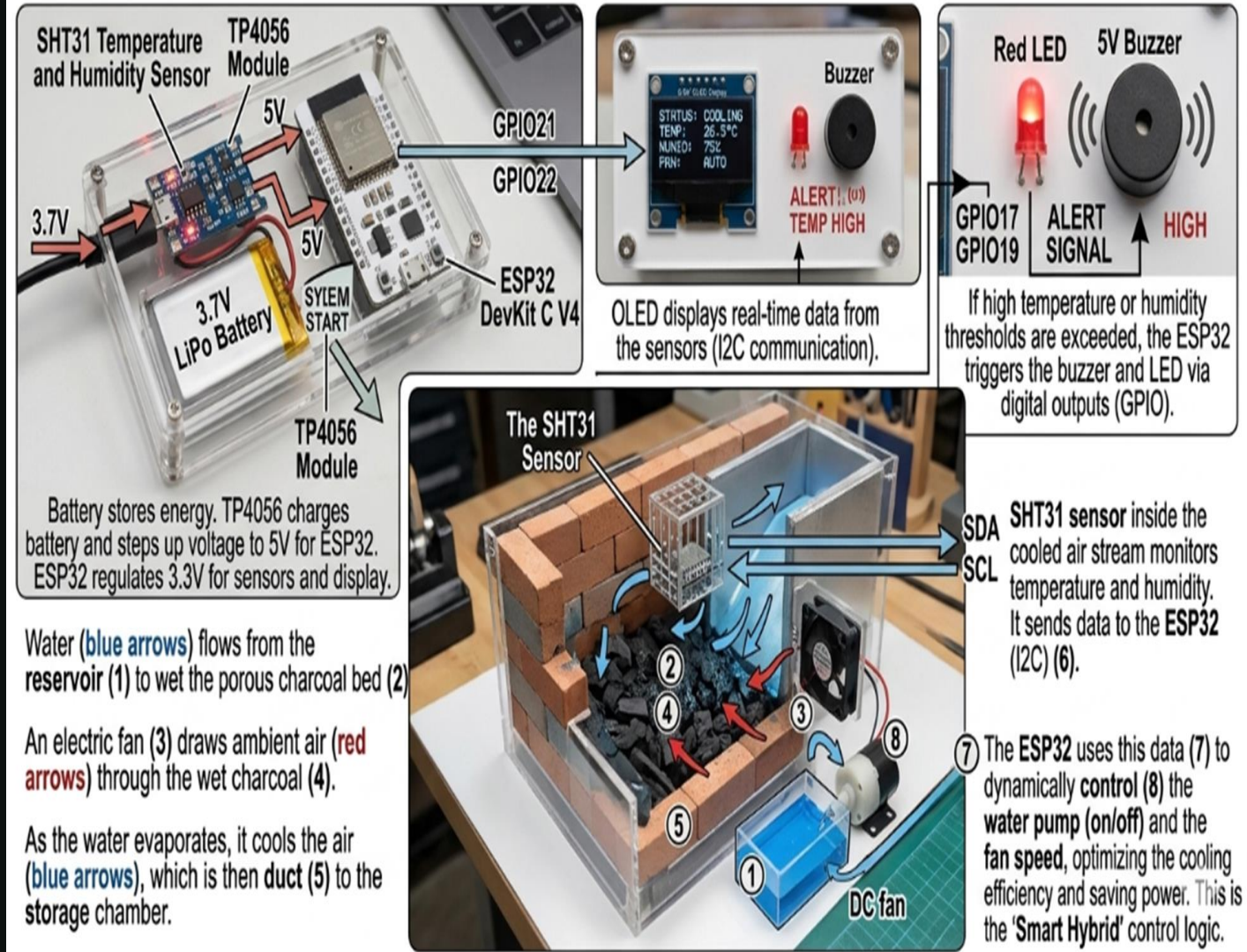


WORKING PRINCIPLE

Evaporative Engineering

1. **Sensing:** I2C SHT3x sensors monitor T/H in real-time.
2. **Cooling:** Solar fans draw air through wet charcoal.
3. **Venting:** Electrochemical sensors track Ethylene gas (C₂H₄). At peak ripening, automated exhaust triggers.

SMART HYBRID CHARCOAL COOLER: PHYSICAL SYSTEM WORKING DIAGRAM

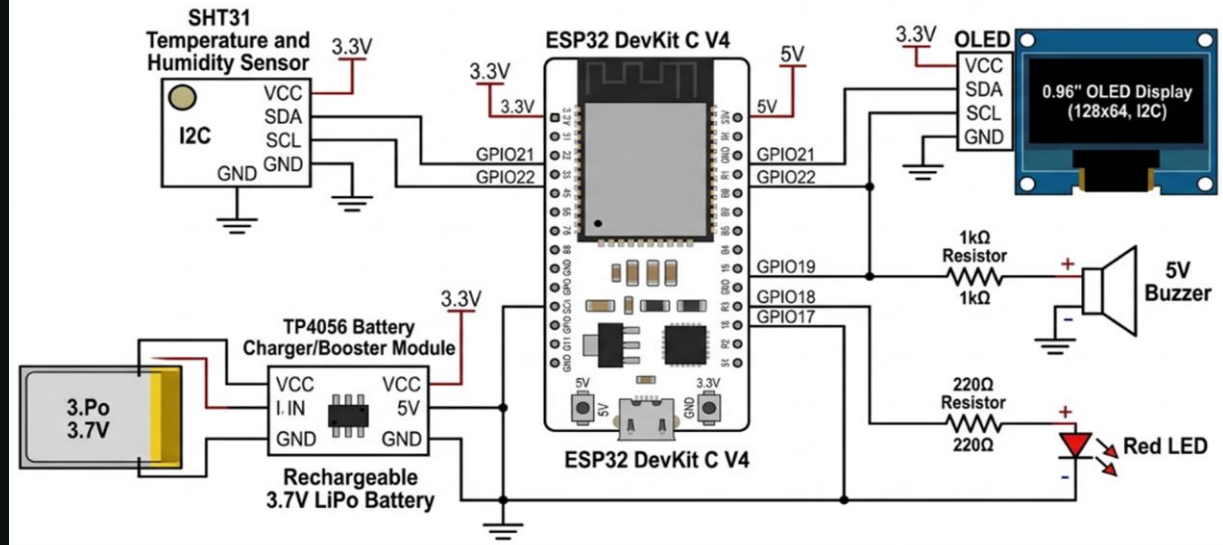


THE DIGITAL BRAIN: ESP32

Full Schematic Control

The system is managed by an **ESP32 DevKit V4**. It integrates high-fidelity sensors and automated relays for zero-human-error storage management.

SMART HYBRID CHARCOAL COOLER: FULL SCHEMATIC DIAGRAM



THE HARVEST HOLD SENSOR SUITE



SHT3x Monitoring

Medical-grade precision for Temperature and Humidity levels, sent directly to the cloud.



Ethylene Sensing

Detects the first signs of ripening gases (C₂H₄) to prevent bulk spoilage through early venting.



LoRa/GSM Sync

Remote data transmission even in the deepest rural hills, ensuring real-time quality audits.

SUSTAINABLE RURAL ENERGY



100% Off-Grid Power

Harvest Hold silos run on **12V Solar PV** systems with 3.7V LiPo battery backup. We use gravity-fed drip systems to minimize water pump energy usage.

- **Low Water Consumption:** < 5L per day.
- **Zero Operating Cost:** Free energy from the Rwandan sun.
- **Resilient Design:** Functions in remote villages.

DIGITAL MARKET MARKETPLACE

AI-Integrated Dashboards

The "Certified Fresh" standard provides a real-time ripening index, allowing buyers to purchase with 100% confidence.

Dashboard for digital agriculture business management

This dashboard providing a visual overview of how agriculture business is performing with a wide range of key performance indicators. It covers aspects like total area, total crop, yield, revenue, weather forecast, task



THE HARVEST HOLD TEAM **current team needed**

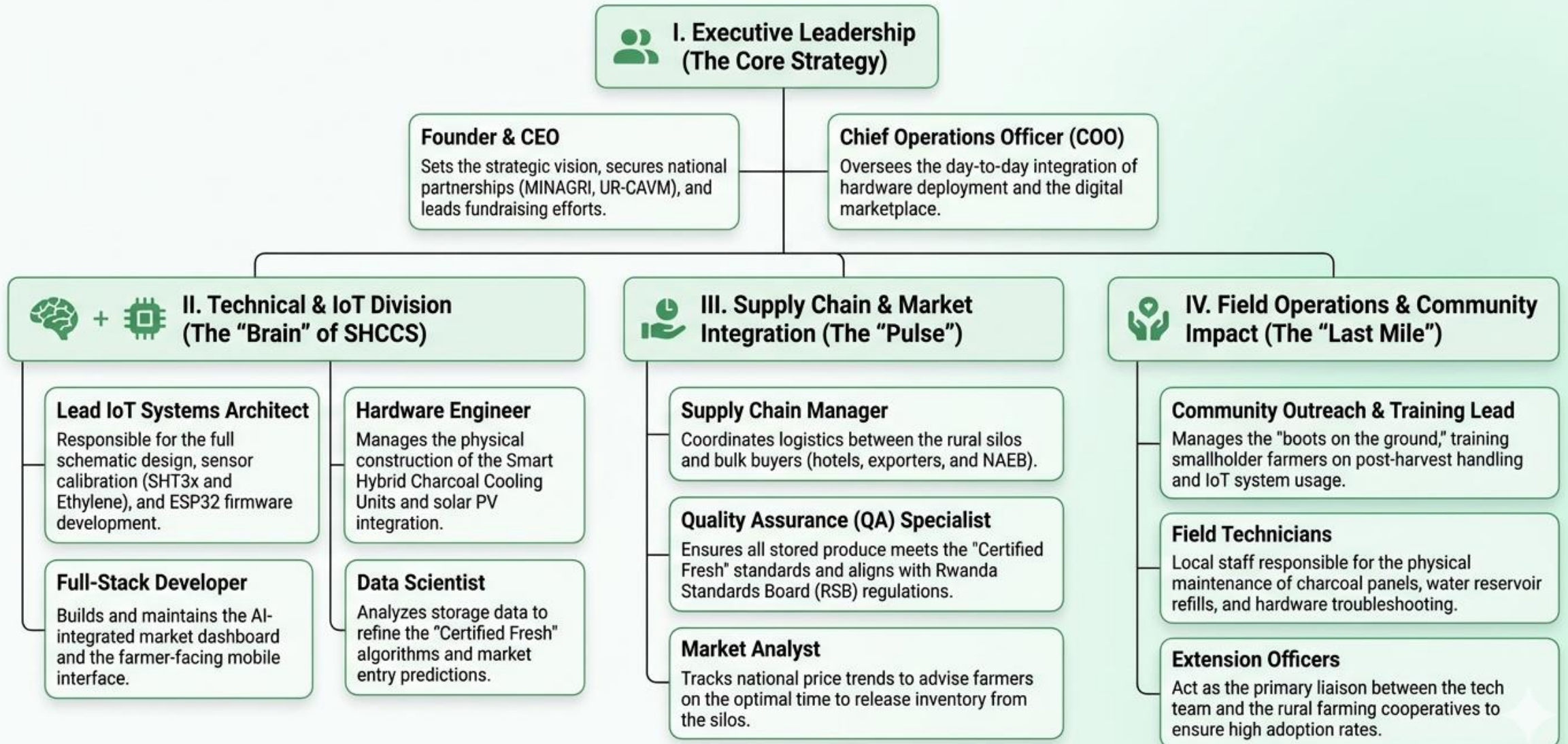
IoT Systems Lead: Sensor architecture & ESP32 development. (developer)

Supply Chain Lead: Marketplace logistics & NAEB partnerships.

Software Developer: AI Dashboard & Mobile Interface.

Community Outreach: Training farmers & coop management. And post harvest management

NEXT ORGANISATION CHAT AFTER SUCCESFULL



STRATEGIC COLLABORATION



MINAGRI

Policy alignment and rural site access for national deployment.



UR-CAVM

Technical R&D, student talent, and lab testing of sensors.



RSB

Standardizing the "Certified Fresh" label for national trust.

NATIONAL SCALING ROADMAP

Q3 2024



Prototype Validation with **UR-CAVM** labs.

Q1 2025



Pilot Deployment (10 Silos) in
Musanze/Bugesera.

Q3 2025



Digital Marketplace Launch &
Exporter onboarding.

2026+



National Rollout: 500+ Silos
across Rwanda.

It is our turn.

Securing Rwanda's Harvest, One Village at a Time.

Questions & Collaboration

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